

Mode 7 — Square

DP16 (16×8) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → DP8 (4×4) → add-and-square

Step 1 — DP16 (16×8)

$$\text{DP16 (16×8)} = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP8 (16×8)

$$\text{DP16 (16×8)} = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 (16×8)}} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 (16×8)}}$$

Step 3 — DP8 (16×4)

$$\text{DP16 (16×8)} = 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 (16×4)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 (16×4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8 (16×4)}} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8 (16×4)}}$$

Step 4 — DP8 (8×4)

$$\begin{aligned} \text{DP16 (16×8)} = & 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hi}}_{\text{DP8 (8×4)}} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lo}}_{\text{DP8 (8×4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hi}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lo}}_{\text{DP8 (8×4)}} \\ & + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hi}}_{\text{DP8 (8×4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{lo}}_{\text{DP8 (8×4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hi}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{lo}}_{\text{DP8 (8×4)}} \end{aligned}$$

Step 5 — DP8 (4×4)

$$\begin{aligned} \text{DP16 (16×8)} = & 2^{16} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{hi}}_{\text{DP8 (4×4)}} + 2^{12} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{hi}}_{\text{DP8 (4×4)}} + 2^{12} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{lo}}_{\text{DP8 (4×4)}} + 2^8 \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{lo}}_{\text{DP8 (4×4)}} \\ & + 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{hi}}_{\text{DP8 (4×4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{hi}}_{\text{DP8 (4×4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{lo}}_{\text{DP8 (4×4)}} + \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{lo}}_{\text{DP8 (4×4)}} \\ & + 2^{16} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{hi}}_{\text{DP8 (4×4)}} + 2^{12} \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{hi}}_{\text{DP8 (4×4)}} + 2^{12} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{lo}}_{\text{DP8 (4×4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{lo}}_{\text{DP8 (4×4)}} \\ & + 2^8 \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{hi}}_{\text{DP8 (4×4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{hi}}_{\text{DP8 (4×4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{lo}}_{\text{DP8 (4×4)}} + \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{lo}}_{\text{DP8 (4×4)}} \end{aligned}$$

Step 6 — Add-and-square (substitute each DP8 (4×4) = $\frac{1}{2} [\sum (x_i + b_i)^2 - \sum x_i^2 - \sum b_i^2]$)

$$\text{DP16 (16×8)} = 2^{16} \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hi})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hi})^2 \right)}_{\text{DP8 (4×4)}}$$

$$\begin{aligned}
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hi})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{lo})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{lo})^2 - \sum_{i=0}^7 (a_{L,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hi})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hi})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{lo})^2 - \sum_{i=0}^7 (a_{H,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{lo})^2 - \sum_{i=0}^7 (a_{L,i}^{lo})^2 - \sum_{i=0}^7 (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hi})^2 - \sum_{i=8}^{15} (a_{H,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{hi})^2 - \sum_{i=8}^{15} (a_{L,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{lo})^2 - \sum_{i=8}^{15} (a_{H,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{lo})^2 - \sum_{i=8}^{15} (a_{L,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{hi})^2 - \sum_{i=8}^{15} (a_{H,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{hi})^2 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{hi})^2 - \sum_{i=8}^{15} (a_{L,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{lo})^2 - \sum_{i=8}^{15} (a_{H,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{lo})^2 - \sum_{i=8}^{15} (a_{L,i}^{lo})^2 - \sum_{i=8}^{15} (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

Step 7 — All components

$$\begin{aligned}
\text{PE} &= 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hi})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{hi})^2 + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{lo})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{hi} + b_i^{lo})^2 \\
&+ 2^8 \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{hi})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{hi})^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{lo} + b_i^{lo})^2 + \sum_{i=0}^7 (a_{L,i}^{lo} + b_i^{lo})^2 \\
&+ 2^{16} \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hi})^2 + 2^{12} \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{hi})^2 + 2^{12} \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{lo})^2 + 2^8 \sum_{i=8}^{15} (a_{L,i}^{hi} + b_i^{lo})^2 \\
&+ 2^8 \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} (a_{H,i}^{lo} + b_i^{lo})^2 + \sum_{i=8}^{15} (a_{L,i}^{lo} + b_i^{lo})^2 \\
\alpha &= 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{hi})^2 + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{hi})^2 \\
&+ 2^8 \sum_{i=0}^7 (a_{H,i}^{lo})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{lo})^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{lo})^2 + \sum_{i=0}^7 (a_{L,i}^{lo})^2 \\
&+ 2^{16} \sum_{i=8}^{15} (a_{H,i}^{hi})^2 + 2^{12} \sum_{i=8}^{15} (a_{L,i}^{hi})^2 + 2^{12} \sum_{i=8}^{15} (a_{H,i}^{hi})^2 + 2^8 \sum_{i=8}^{15} (a_{L,i}^{hi})^2 \\
&+ 2^8 \sum_{i=8}^{15} (a_{H,i}^{lo})^2 + 2^4 \sum_{i=8}^{15} (a_{L,i}^{lo})^2 + 2^4 \sum_{i=8}^{15} (a_{H,i}^{lo})^2 + \sum_{i=8}^{15} (a_{L,i}^{lo})^2 \\
\beta &= 2^{16} \sum_{i=0}^7 (b_i^{hi})^2 + 2^{12} \sum_{i=0}^7 (b_i^{hi})^2 + 2^{12} \sum_{i=0}^7 (b_i^{lo})^2 + 2^8 \sum_{i=0}^7 (b_i^{lo})^2 \\
&+ 2^8 \sum_{i=0}^7 (b_i^{hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{lo})^2 + \sum_{i=0}^7 (b_i^{lo})^2 \\
&+ 2^{16} \sum_{i=8}^{15} (b_i^{hi})^2 + 2^{12} \sum_{i=8}^{15} (b_i^{hi})^2 + 2^{12} \sum_{i=8}^{15} (b_i^{lo})^2 + 2^8 \sum_{i=8}^{15} (b_i^{lo})^2 \\
&+ 2^8 \sum_{i=8}^{15} (b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} (b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} (b_i^{lo})^2 + \sum_{i=8}^{15} (b_i^{lo})^2
\end{aligned}$$